

Bayesian Project, Enzyme Kinetics

Group 11

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1. Introduction

Enzyme kinetics is a field in biochemistry concerned with the rates of chemical reactions catalyzed by enzymes, which is crucial, since these rates form the basis for metabolic research, drug development, and industrial biotechnology. Through this report, we analyse the Puromycin dataset in R, which records the reaction velocity of an enzymatic process at different substrate concentrations under two experimental conditions: treated with Puromycin and untreated.

The primary objective here is to quantify how the presence of the antibiotic Puromycin affects the relationship between substrate concentration and reaction rate. In this report, we employ a Bayesian nonlinear regression framework to estimate the parameters of the Michaelis–Menten model, allowing us to account for uncertainty and provide probabilistic comparisons between the two treatment groups.

1.2 Michaelis–Menten Model

The Michaelis–Menten equation is a baseline model used for describing the kinetics of enzymes by relating the initial reaction rate(V) to the substrate concentration(S) and is defined as follows.

$$V = \frac{V_{max}(S)}{K + S}$$

where:

- V (dependent variable) represents the velocity of the reaction
- S (independent variable) represents the substrate concentration
- V_{max} represents the maximum rate at which the enzyme functions when fully saturated with the substrate
- K (Michaelis Constant) is the substrate concentration at which the reaction rate is exactly half of V_{max}

A lower K implies the enzyme reaches its maximum potential at lower concentrations, indicating higher efficiency or affinity.

We treat the parameters as random variables within a Bayesian context, and the model is formulated as follows:

$$Y_i \sim N(\mu_i, \sigma^2) \text{ where } \mu_i = \frac{V_{max,j} x_i}{K_j + x_i}$$

2. Methodology

To estimate the parameters, the JAGS (Just Another Gibbs Sampler) software was used via the rjags interface to find point estimates, credibility intervals for the parameters, and derive the posterior distributions for quantities required for reporting. These include the ratio of velocities between groups and the probability that the treated group outperforms the untreated group.

As specified in the model, the observed reaction rates Y_i are normally distributed around μ_i with error σ^2 , meaning we employ a Normal Likelihood. However, the parameters V_{max} and K are nonlinear in the mean structure, making the use of Markov Chain Monte Carlo methods necessary to sample from the posterior distribution, as there is no analytical solution for the estimates. Additionally, we utilize uninformative priors to ensure that results are driven by the experimental data rather than preconceived biases, while enforcing strictly positive constraints ($V_{max}, K > 0$) to align with reality.

3. Questions

1. Write out BUGS code for a Bayesian nonlinear regression model based on the Michaelis–Menten equation. Assume uninformative priors for all the parameters. Explain your code, indicating what is the likelihood, the nonlinear mean structure, the prior distributions, and the role of each parameter.

Provided data is a list with variables for the reaction rate (Y), the concentration (conc), the group index (j), the total number of observations (N), and the ratios specified in the 4th question.

The BUGS code is as follows:

```
"
model {
  for (i in 1:N) {
    rate[i] ~ dnorm(mu[i], tau)
    mu[i] <- Vmax[group[i]] * conc[i] / (K[group[i]] + conc[i])
  }

  for (j in 1:2) {
    Vmax[j] ~ dnorm(0, 0.0001) I(0, )
    K[j] ~ dunif(0, 2)
  }

  tau ~ dgamma(0.0001, 0.0001)
  sigma <- 1 / sqrt(tau)

  ratio_Vmax <- Vmax[1] / Vmax[2]
  ratio_K <- K[1] / K[2]
}
"
```

As the likelihood represents the probability of the observed data given the parameters, here we assume the reaction rates ($rate[i]$) are independent and normally distributed ($rate[i] \sim N(\mu[i], \tau)$). Since BUGS uses precision ($\frac{1}{\sigma^2}$) instead of variance, tau has been explicitly declared within the function as a proxy for variance. The mean is no longer a linear combination of parameters. Instead, it follows the functional form defined by the Michaelis-Menten equation, as it accounts for the saturation effect that occurs when reaction rates slow down as concentration increases.

The parameter $Vmax[j]$ (maximum velocity) represents the limit of the reaction rate as concentration x approaches infinity and signifies the total capacity of the enzyme system. The Michaelis constant K is the concentration at which the reaction rate is half of $Vmax$ and serves as an inverse measure of the enzyme's affinity for the substrate. Therefore, a smaller K indicates the enzyme reaches high speeds at lower concentrations. The residual standard deviation σ represents measurement error in the enzymatic process.

As we have no strong previous knowledge, we use uninformative priors to let the experimental data decide the results of the model. Since $Vmax$ is the maximum possible reaction rate, and reaction rate is specified by the model to be normally distributed, we chose a normal prior with a large variance (small precision), while truncating at zero to ensure the values remain strictly positive. On the other hand, since the distribution for K is not as easily obtained (given the non-linear nature of the mean function), we chose a uniform distribution as a non-informative prior for K , as it requires the fewest amount of assumptions. The lower and higher bounds of the uniform distribution are specified based on the initial data exploration done at the beginning of the R code, to make sure that all the possible values of K are contained in the parameter space. For tau, the prior is a flat Gamma distribution, which ensures that the variance remains positive and poorly defined initially.

2. Run the MCMC algorithm. Describe your choices (number of chains, iterations, burn-in). Assess convergence of the chains. Use appropriate diagnostics and summarize your conclusions about convergence.

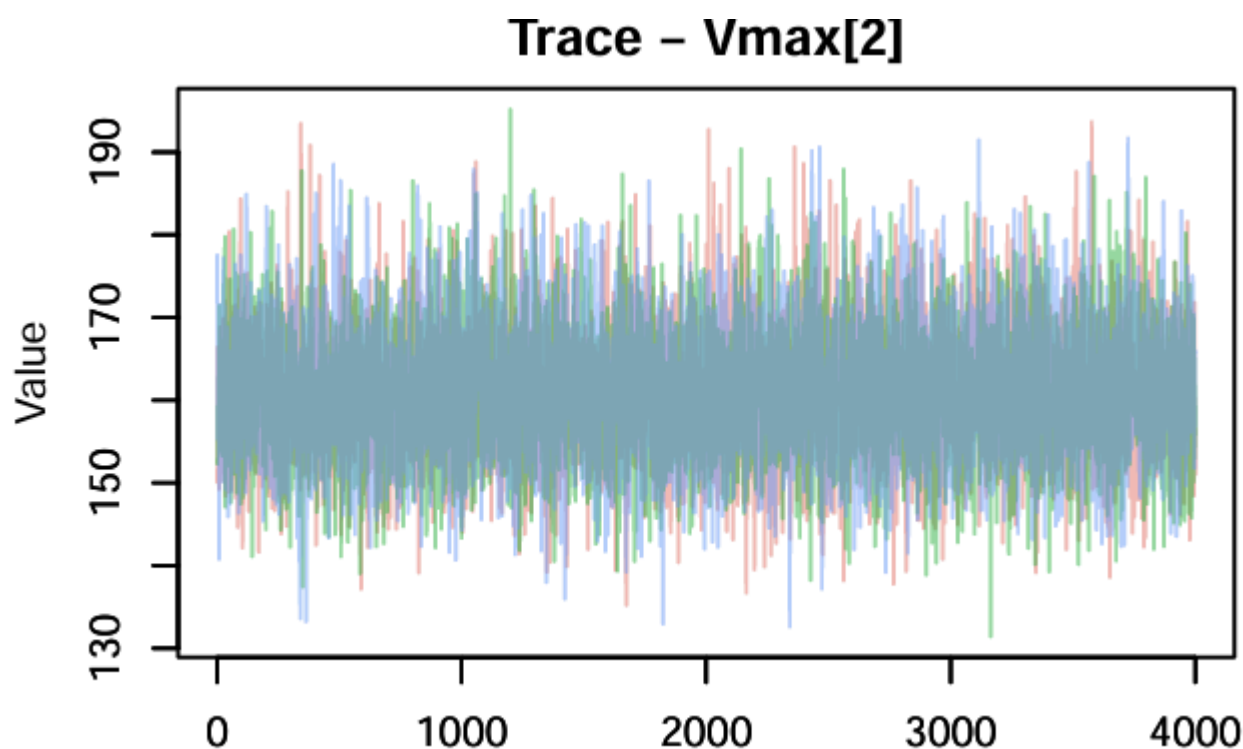
The MCMC algorithm was run with 3 chains for 20,000 iterations, with a burn-in of 5,000 iterations. Convergence was assessed using the Gelman-Rubin diagnostic (R-hat). For all parameters, the R-hat value is 1.00, which is well below the common threshold of 1.1, indicating the chains have successfully mixed and reached a stationary distribution.

We chose to use 3 chains starting from different values to ensure that the posterior space is explored thoroughly enough to prevent the results from depending on a single starting point, without increasing complexity. Having three chains provides a tie-breaker, in case one chain does not converge, and sufficient evidence of stability while keeping the traceplots easy to interpret.

Setting iterations to 20,000 per chain ensures that the distribution of samples accurately approximates the true posterior distribution, whereas a burn-in of the first 5,000 samples helps eliminate the influence of the initial starting value bias. This allows the chains to reach their stationary distribution before samples are recorded, and with a setup involving 3 distinct chains, this makes the final samples more stable.

Convergence Assessment was performed using three primary diagnostics:

- **Gelman-Rubin Diagnostic:** The R-hat compares the variance within chains to the variance between chains, and since all R-hat values here were below the critical threshold of 1.1, we have strong evidence that the chains have successfully converged.
- **Effective Sample Size:** The maximum possible effective size in our setting is 12,000 (4,000 samples per chain since thinning was set to 5), and the effective sample sizes of our parameters are relatively close to the maximum, showing that our model parameters are well-specified, and that thinning was effective in addressing autocorrelation.
- **Trace Plots:** The Traceplots of all 3 chains overlap without any visible signs of drift, confirming convergence visually. (see the PDF file MCMC_Diagnostics for complete visualizations)



3. Give (point and interval) posterior summaries for all model parameters. Interpret the parameters.

Parameter	Posterior Mean	95% HPD Interval	95% Equal-tail Interval	Interpretation
V_{max}[1] (Treated)	212.470	[197.825, 226.699]	[198.198, 227.325]	Maximum velocity of treated enzyme
V_{max}[2] (Untreated)	160.419	[145.574, 176.105]	[145.953, 176.562]	Maximum velocity of untreated enzyme
K[1] (Treated)	0.064	[0.048, 0.083]	[0.049, 0.084]	Substrate concentration to reach 50% V _{max} [1]
K[2] (Untreated)	0.049	[0.032, 0.070]	[0.032, 0.071]	Substrate concentration to reach 50% V _{max} [2].
sigma	10.605	[7.4, 14.605]	[7.880, 15.410]	standard deviation

V_{max} of the treated group (212.470) is significantly higher when compared to the untreated group (160.419), and neither of their 95% Credible Intervals overlaps, indicating that the treatment increases the enzyme's processing capacity.

Since K is an indicator of the enzyme's affinity for its substrate, lower K values suggest higher affinity. Here, the untreated group has a slightly lower mean K (0.049) than the treated group (0.064), but the noticeable overlap in their Credible Intervals suggests that the treatment's effect on affinity (K) might not be statistically significant.

A posterior mean variance of 10.605 indicates that the observed reaction rates typically deviate from the predicted curve by about 10.6 units, maybe due to measurement error.

4. Calculate the ratio $V_{max,treated}/V_{max,untreated}$ and $K_{max,treated}/K_{max,untreated}$. Compute posterior summaries for these ratios. Interpret the results.

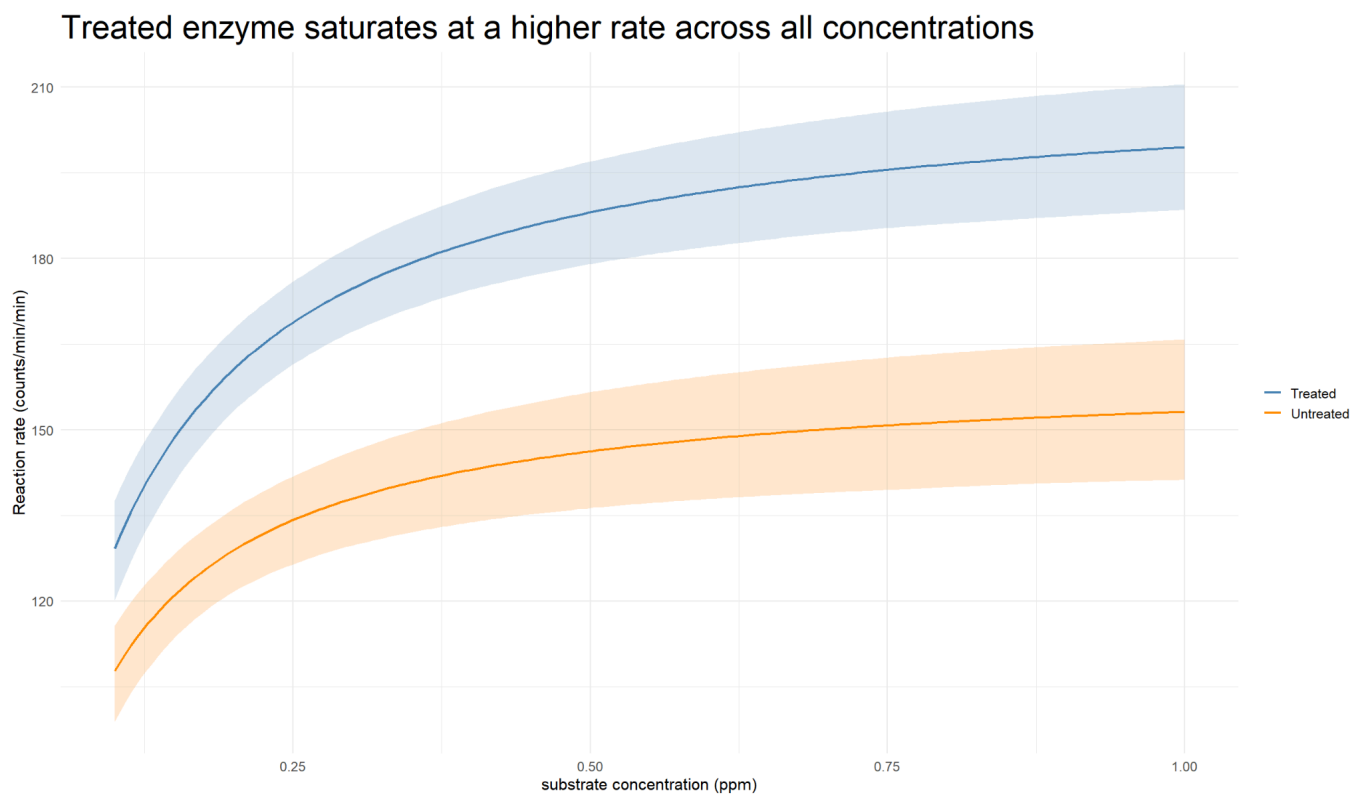
Ratio	Posterior Mean	95% HPD Interval	95% Equal-tail Interval
$\frac{V_{max[treated]}}{V_{max[untreated]}}$	1.3	[1.17, 1.5]	[1.18, 1.5]

$\frac{K_{treated}}{K_{untreated}}$	1.4	[0.75, 2.0]	[0.82, 2.2]
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The Vmax ratio shows that the treated group, on average, has a 30% higher rate than the untreated, and the CIs show that the treatment significantly increases the maximum velocity of the reaction since the interval does not contain 1. Also, the HPD and equal-tail intervals for this parameter are very similar, indicating symmetry in distribution.

Meanwhile, the K ratio compares the half-saturation constants, and here it can be seen that the treated group needs 40% more substrate to reach Vmax when compared to the untreated group. However, the 95% credible intervals for this ratio are very wide and include 1, meaning there is no statistical significance in the finding that the treatment alters the affinity of the enzyme for the substrate, and the difference could be due to noisy data.

5. Make a plot of the fitted reaction curves for both treatment groups, for concentrations between 0.1 and 1, including uncertainty bands. Interpret the results.



The vertical separation between the two curves, with the treated group (blue) curve plateauing at a higher level than the untreated group (orange), visually confirms our posterior summary of the Vmax ratio. Treatment with the antibiotic seems to significantly increase the maximum capacity of the reaction. The shaded regions represent the 95% Credible Intervals for the mean reaction rate, and the fact that they do not overlap at any point across the concentration range is evidence that the difference in reaction rates is statistically significant, meaning that the treated enzyme performs better than the untreated one at every tested concentration level.

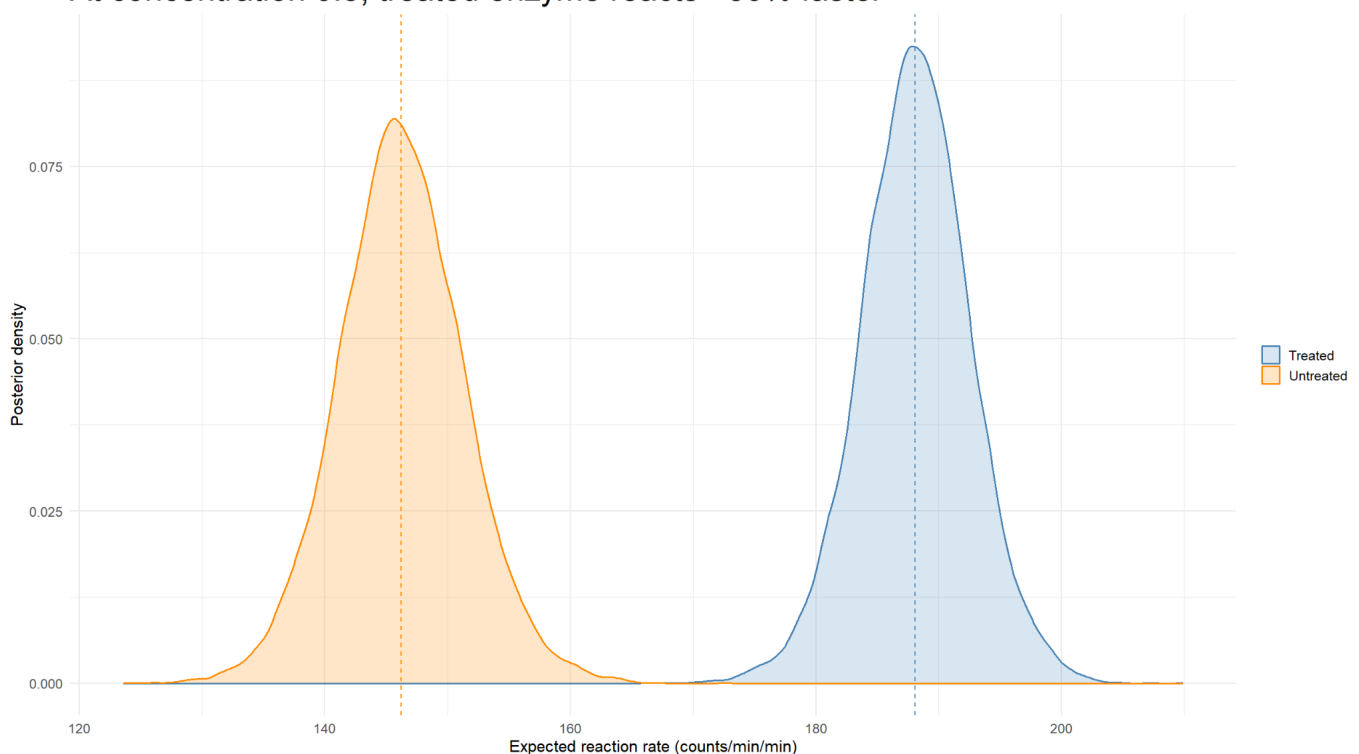
The Michaelis constant determines the steepness of the curve, and both show a similar sharp rise at low concentrations (< 0.25) before leveling off. The similar steepness in curves indicates that while treatment has a higher asymptote, affinity might not be fundamentally changed by treatment. This finding also aligns with our analysis of the K ratio, which was not statistically significant.

6. Compute the posterior distribution of the reaction rate at concentration $x = 0.5$ for both treatment groups. Make posterior density plots and interpret the results.

Group	Posterior Mean	95% HPD Interval	95% Equal-tail Interval
Treated	188.1	[179.3, 197.1]	[179.1, 196.9]
Untreated	146.1	[135.9, 156.1]	[136.0, 156.2]

At a substrate concentration of 0.5, there is a clear separation (188.1 vs 146.1) between these two posterior distributions, indicating a robust treatment effect as highlighted by the image below.

At concentration 0.5, treated enzyme reacts ~30% faster



While the previous plot showed the trend across all concentrations, this one focuses on one specific point and, by doing so, shows the complete separation between the distributions. It can be seen that the two density distributions (Treated (blue) and Untreated (orange)) do not overlap at all, which means that, as per our data, there is a very high probability that the reaction rate is higher for the treated group than the untreated group at this specific concentration. The peaks of the curves (modes) represent the most likely reaction rates, and it is clearly visible that the treated group has a higher value than the untreated at 0.5 ppm concentration. Both bell curves have similar widths, indicating that uncertainty regarding reaction rates is comparable for both.

7. Estimate the posterior probability that the maximum reaction rate is higher for the treated group than for the untreated group: $P(V_{\max, \text{treated}} > V_{\max, \text{untreated}} \mid \text{data})$.

Estimate the posterior probability that the half-saturation constant is smaller for the treated group than for the untreated group: $P(K_{\text{treated}} < K_{\text{untreated}} \mid \text{data})$.

Explain how these probabilities are obtained. Interpret the result.

Hypothesis	Estimated Posterior Probability
$P(V_{\max [\text{treated}]} > V_{\max [\text{untreated}]} \mid \text{data})$	1.0000 (100%)
$P(K_{\text{treated}} < K_{\text{untreated}} \mid \text{data})$	0.1137 (11.37%)

These are calculated using the proportion of successful iterations from the MCMC chains.

In every iteration, we check if $V_{\max [\text{treated}]} > V_{\max [\text{untreated}]}$ and count as 1 if true and 0 otherwise. Once we have gone through every sample, we take the average, which gives us the probability. Similarly, we count how many times K_{treated} was smaller than $K_{\text{untreated}}$ using the same method.

The probability of 1 indicates that in every one of our 4,000 samples, the treated group had a higher maximum velocity than the untreated group. Thus, we state with absolute certainty (given this model and data) that the Puromycin treatment increases the total capacity of the enzymatic reaction.

The probability of 0.1137 (11.37%) shows there is only an 11.37% chance that the treated enzyme is more efficient than the untreated one, and an almost 89% probability that the treated group is less efficient. However, through earlier analysis, we have seen there is not a lot of statistical evidence for the effect of treatment on affinity, so this must be reported with caution.

8. Determine the concentration at which the expected reaction rate reaches 80% of the maximum: $\mu(x) = 0.8 V_{\max}$. Compute the posterior distribution of this quantity for both treatment groups and interpret the results.

To find the concentration at which the reaction rate reaches **80% of $V_{\max}$** , we can solve the Michaelis–Menten equation for x when $\mu(x) = 0.8 V_{\max}$ as follows

$$0.8V_{\max} = \frac{V_{\max} * x}{K + x}$$

On solving, we see that the concentration required to reach 80% capacity is **four times** the Michaelis constant.

Posterior Summary for 80% Capacity Concentration

Group	Posterior Mean (x)	95% HPD Interval	95% Equal-tail Interval
Treated ($x_{80_{\text{treated}}}$)	0.259	[0.191, 0.332]	[0.195, 0.337]
Untreated ($x_{80_{\text{untreated}}}$)	0.197	[0.127, 0.28]	[0.129, 0.283]

The treated group requires a concentration of approximately **0.259** to reach 80% of its maximum velocity, while the untreated group reaches it at **0.197**, reflecting the enzyme's affinity. Since the treated group requires a higher concentration, it suggests that the treatment might slightly decrease the enzyme's affinity for the substrate, although we saw earlier that this difference is not statistically significant at the 95% level (credibility intervals for $x_{80_{\text{treated}}}$ and $x_{80_{\text{untreated}}}$ overlap here as well). However, purely based on these results, we see that while the treated enzyme has a higher overall capacity, it requires a higher substrate concentration to reach it compared to the untreated enzyme.